

GAGNOA, Côte d'Ivoire

WMO: 655570

Lat: 6.133N

Lon: 5.950W

Elev: 689

StdP: 14.33

Time Zone: 0.00 (E00)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	63.8	67.7	52.2	59.4	84.9	57.7	72.9	76.5	10.0	79.0	9.5	78.8	0.3	0	0.255

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	18.1	95.0	76.7	93.5	77.3	92.6	77.5	81.0	90.2	80.4	89.8	79.8	89.0	3.9	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.8	153.3	86.5	77.8	148.4	85.1	77.3	145.7	84.3	45.4	90.7	44.6	90.1	44.0	89.7	86.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.4	7.8	7.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	80.8	81.3	83.3	83.0	82.7	81.6	79.8	78.2	77.8	79.8	80.8	81.2	80.6
	DBStd	2.61	3.28	2.47	2.10	1.98	2.13	1.74	1.63	1.59	1.61	1.52	1.33	2.11
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11248	971	932	1023	980	979	893	875	863	894	954	937	947
	CDD65	5773	506	512	558	530	514	443	410	398	444	489	487	482
	CDH74	49604	5004	5292	5622	5367	4728	3271	2645	2427	3165	3720	4085	4278
	CDH80	19530	2339	2561	2571	2361	1839	994	648	562	1019	1302	1601	1733
(14) Wind	WSAvg	2.6	2.4	2.4	2.5	2.5	2.5	2.6	2.8	3.3	3.1	2.8	2.8	1.9
(15) Precipitation	PrecAvg	55.0	0.8	2.6	4.9	7.1	7.2	7.9	3.5	3.5	6.2	6.3	4.0	1.6
	PrecMax	78.7	5.9	6.2	10.7	12.8	14.0	17.4	15.7	10.7	13.8	12.6	9.2	6.8
	PrecMin	37.6	0.0	0.1	0.6	3.3	3.1	1.8	0.3	0.4	0.6	2.6	0.5	0.0
	PrecStd	9.0	1.2	1.5	2.1	2.2	2.4	3.3	2.7	2.2	3.3	2.4	2.3	1.7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	95.4	97.2	96.1	94.8	93.2	90.0	88.0	87.8	89.9	91.4	92.0	93.1
		MCWB	73.5	74.6	78.7	79.9	80.8	79.1	77.2	76.3	77.4	78.6	78.7	74.1
	2%	DB	93.6	95.3	94.3	93.3	91.4	88.2	86.3	86.2	88.4	89.8	90.5	91.2
		MCWB	74.0	76.0	78.9	79.9	80.0	78.6	76.7	76.3	77.3	78.0	78.4	75.4
	5%	DB	92.3	93.7	92.9	91.8	89.9	86.9	85.3	84.7	87.3	88.4	89.5	89.7
		MCWB	74.3	76.5	78.7	79.5	79.5	78.2	76.3	76.1	77.1	77.9	78.3	75.8
	10%	DB	90.5	92.1	91.3	90.5	88.5	85.5	83.8	83.5	85.7	87.0	88.0	88.2
		MCWB	74.3	76.6	78.6	79.2	79.1	77.7	75.9	75.8	76.8	77.7	78.1	76.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	79.2	80.9	81.6	82.1	81.9	80.7	79.1	78.2	79.1	79.8	80.5	79.7
		MCDB	89.2	92.2	91.8	91.0	90.1	87.5	85.6	84.7	87.1	87.7	88.9	87.6
	2%	WB	78.3	79.8	80.6	80.9	80.9	79.5	77.8	77.3	78.3	79.0	79.6	78.9
		MCDB	88.0	90.7	91.1	90.8	89.4	86.4	84.5	84.0	86.2	87.3	88.0	87.3
	5%	WB	77.7	79.0	79.8	80.1	80.1	78.7	77.0	76.6	77.6	78.4	79.0	78.2
		MCDB	86.9	89.5	90.1	89.8	88.3	85.5	83.5	83.2	85.2	86.5	87.4	86.2
	10%	WB	77.1	78.2	79.1	79.4	79.3	77.9	76.3	76.0	77.0	77.9	78.4	77.6
		MCDB	85.7	87.6	89.2	88.6	87.1	84.1	82.2	82.1	84.4	85.6	86.5	85.0
(35) Mean Daily Temperature Range	5% DB	MDBR	20.6	19.8	18.1	17.1	15.3	13.1	12.2	12.0	13.7	14.6	15.8	17.8
		MCDBR	21.7	21.5	19.8	18.5	16.8	14.8	14.0	13.9	15.3	16.0	16.9	19.3
		MCWBR	6.3	6.1	7.0	6.6	6.7	6.4	5.5	5.6	5.5	5.7	6.5	6.7
	5% WB	MCDBR	18.2	19.1	18.7	17.9	16.1	13.9	12.7	12.6	14.3	15.3	16.3	16.4
		MCWBR	5.7	6.3	7.4	7.1	6.9	6.5	5.6	5.7	5.8	6.0	6.7	6.1
(40) Clear-Sky Solar Irradiance	taub		0.738	0.720	0.748	0.587	0.507	0.486	0.478	0.461	0.454	0.479	0.544	0.660
	taud		1.586	1.651	1.608	1.944	2.148	2.214	2.238	2.291	2.322	2.243	2.010	1.740
	Ebn at Noon		201	210	206	239	252	254	258	267	273	267	246	215
	Edh at Noon		87	83	87	61	49	45	44	43	42	45	56	73
(44) All-Sky Solar Radiation	RadAvg		1529	1522	1625	1653	1536	1293	1231	1200	1306	1427	1457	1426
	RadStd		74	57	67	66	58	66	61	74	54	48	47	46

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page